Code No. 3420

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE, AI&ML) IV-Semester (MAIN) Examinations Jun/Jul-2026

SIGNALS AND SYSTEMS

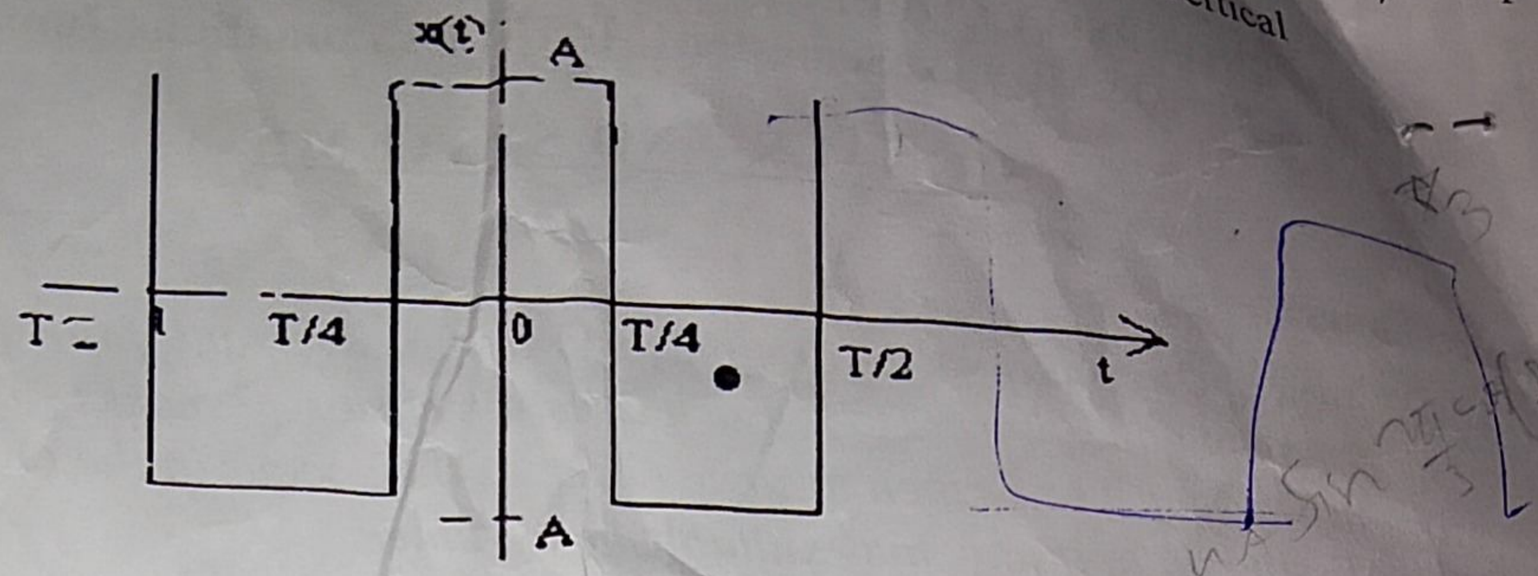
Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1 a)	Test whether the signal $x(t)=e^{-t}$ is even, odd or neither.	2	4	1
b)	Find the fundamental period of the signal $x(t) = \sin(4\pi t)$	2	1	2
c)	Determine the inverse Laplace Transform of $X(s) = \frac{1}{S+2}$	2	5	3
d)	Find the fundamental period of the discrete-time signal $x(n) = \cos(\frac{\pi n}{4})$	2	2	4
e)	Find the DTFT of $x(n) = u(n)$	2	2	5
f)	Find the Z-Transform of $a^n u(n)$	2	1	5
2. a)	Analyze whether the following systems are causal, stable, linear and time invariant i) $y(t) = x(t)+x(t-2)$ ii) $y(t) = tx(t)$	6	4	1
b)	Perform time shifting, time scaling and time inversion on the signal $x(t) = u(t-1)$ $x(t) = u(t-1)$ and sketch the resulting waveforms..	6	4	1
3. a)	Derive the Trigonometric Fourier Series representation for periodic signals.	5	5	2

{ P.T.O.}

4. a) Find Trigonometric Fourier series for a periodic square waveform shown in figure which is symmetrical with respect to the vertical axis. 7 1 2



b) Verify the time shifting and frequency shifting properties of Fourier Transform for suitable signals. 6 2 3

5. a) Derive the Fourier Transform of the rectangular pulse signal. 6 3 3

b) Derive the Discrete-Time Fourier Series (DTFS) coefficients for a periodic sequence. 6 4 4

6. a) Perform discrete-time signal operations such as shifting, folding and scaling on $x(n)= \{1,2,3,4,5,8\}$ 6 5 4

b) Solve the difference equation using Z-Transform: 6 4 5

$$y[n] - 0.5y[n-1] = x[n]$$

7. a) State and prove the following properties of z-transform: 6 1 5
(i) Time reversal (ii) Conjugation

b) Find whether the signal are energy signal or power signal 6 2 1

i) u(t)+u(t-2) ii) $e^{-5t}u(t)$

b) Explain the properties of the region of convergence of X (z). 6 3 5

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